GANNAVARAPU GOKUL MADHAV

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SUMMARY

Aspiring AI/ML Engineer and Full-Stack Developer with experience in web/mobile apps and machine learning. Proficient in Java, Python, JavaScript, and algorithms, with a passion for solving real-world problems.

SKILLS

Java, Python, JavaScript, HTML5, CSS, PHP, YOLOv5, Supervised Learning, PyTorch, MySQL, Git, Flask, AI Product Delivery, Business Analysis, Project Management

EXPERIENCE

RESEARCH INTERN — NIT ROURKELA *(July 2025 – August 2025)*

- Worked on a 3D face reconstruction project using DECA and Poisson Surface Reconstruction techniques
- Implemented RANSAC-based landmark refinement and generated high-fidelity 3D meshes from 2D facial images.

EDUCATION

B.TECH in CSE(AI&ML) — LAKIREDDY BALI REDDY COLLEGE OF ENGINNERING *(2026)*

Diploma in CSE — SREE VAHINI INSTITUTE OF SCIENCE AND TECHNOLOGY *(2023)*

SSC — GOWTHAM E.M HIGH SCHOOL *(2020)*

PROJECTS

Mosquito Classification using YOLOv5

- Developed an AI model to classify mosquito species with ~90% accuracy using YOLOv5, CSPDarknet53 (feature extraction), and PANet (neck structure analysis).
- Trained on a custom dataset of 1,000+ images, optimizing hyperparameters for precision.

Technologies: Python, Deep Learning

Real-Time Brain Tumor Classification from MRI Scans

- Developed a lightweight 3D CNN model using EfficientNet-Lite for real-time brain tumor classification on the BraTS 2021 dataset.
- Optimized model with pruning, quantization, and TensorRT, achieving 87% Dice score and 18 FPS on NVIDIA RTX 4060 GPU.

Technologies: Python, EfficientNet-Lite 3D, Deep Learning

Skin Disease Classification using HAM10000 Dataset

- Implemented an ensemble of EfficientNet-B5, ResNet-101, and a custom CNN to classify 7 skin lesion types into 3 broad categories (Cancerous, Benign, Vascular).
- Achieved 89% test accuracy using transfer learning and data augmentation on the HAM10000 dataset.

Technologies: Python, CNN, EfficientNet-B5, ResNet-101

CRACKERWALA – E-COMMERCE PLATFORM FOR CRACKERS

- Built a PHP-based shopping website (crackerwala.in) with product listings, cart, and payment integration.
- Designed MySQL database for inventory/order management and jQuery for dynamic UI.
- Reduced checkout time by 30% with optimized SQL queries.

Technologies: PHP, MySQL, JavaScript

EDUMATE – EDUCATIONAL VIDEO STREAMING PLATFORM

- Created a video-on-demand platform for educational content with user subscriptions and progress tracking.
- Integrated PHP backend for video uploads/streaming.

Technologies: PHP, HTML5

3D FACE RECONSTRUCTION FROM 2D IMAGES

- Developed a 3D face reconstruction pipeline using DECA (Detailed Expression Capture and Animation) model from single 2D images.
- Integrated RANSAC-based 3D landmark refinement and Poisson surface reconstruction for high-quality mesh generation.

Technologies: Python, PyTorch, OpenCV, DECA